WEEK 03 — Number Systems & Number-System Conversions

Complete Notes
English Only
Numerical Focus

This week is extremely important because it contains many numerical questions. You should understand the conversion methods rather than memorize answers.

According to your syllabus, Week 3 covers:

1. Decimal, Binary, Octal and Hexadecimal Number Systems.
2. Positional Notation.
3. Representation of Numbers in Computers.
4. Decimal-to-Binary Conversion.
5. Binary-to-Decimal Conversion.
6. Binary-to-Hexadecimal Conversion.
7. Hexadecimal-to-Binary Conversion.
8. Octal Conversions.
9. Conversion Practice.

PART 1: Introduction to Number Systems

1. What is a Number System?

Definition:

A number system is a method of representing numbers using a specific set of digits or symbols according to certain rules.

Explanation

Computers use different number systems to represent, store, and process data.

Humans normally use the decimal number system, while computers internally represent information using binary digits (0 and 1).

There are four major number systems used in Computer Organization:

1. Decimal Number System
2. Binary Number System
3. Octal Number System

4. Hexadecimal Number System

2. Important Terms

Before studying the number systems, understand these basic terms.

A. Digit

A digit is a symbol used to represent a number.

Examples: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9.

B. Base (Radix)

The base of a number system represents the total number of unique digits used in that system, including zero.

Examples:

- Decimal has base 10.
- Binary has base 2.
- Octal has base 8.
- Hexadecimal has base 16.

C. Bit

A bit is the smallest unit of digital information and can have only two values: 0 or 1.

D. Nibble

A nibble is a group of 4 bits.

Example:

\[1010 = 1\ Nibble \]

E. Byte

A byte consists of 8 bits.

Example:

\[10101010 = 1\ Byte \]

PART 2: Types of Number Systems

3. Decimal Number System (Base 10)

PPT - Binary Numbers PowerPoint Presentation, free download - ID:3038174
slideserve.com

Definition:

The Decimal Number System is a number system with base 10 that uses ten digits, from 0 to 9, to represent numbers.

Digits Used

$$ 0,1,2,3,4,5,6,7,8,9 $$

Characteristics

1. It has base 10.
2. It uses ten unique digits.
3. It is commonly used in daily life.
4. Each digit has a positional value.
5. Each position represents a power of 10.

Example

Consider:

$$ (253)_{10} $$

Expand it using positional notation:

$$ =2\times10^2+5\times10^1+3\times10^0 $$
$$ =200+50+3 $$
$$ =253 $$

Important: The subscript 10 represents the base of the number.

4. Binary Number System (Base 2)

Data representation UNIT-1 | PPT
slideshare.net

Definition:

The Binary Number System is a number system with base 2 that uses only two digits, 0 and 1, to represent all numbers.

Digits Used

$$ 0,1 $$

Characteristics

1. It has base 2.
2. It uses only two digits.
3. It is the fundamental number system of digital computers.
4. Each position represents a power of 2.
5. It is used to represent data and instructions inside computers.

Why Do Computers Use Binary?

Computers use electronic circuits that can represent two stable states:

- ON = 1
- OFF = 0

These two states make binary representation suitable for digital electronic systems.

Example

Consider:

$$ (1011)_2 $$

Expand:

$$ =1\times2^3+0\times2^2+1\times2^1+1\times2^0 $$
$$ =8+0+2+1 $$
$$ =(11)_{10} $$

5. Octal Number System (Base 8)

Number Systems
krobbins.com

Definition:

The Octal Number System is a number system with base 8 that uses eight digits, from 0 to 7.

Digits Used

$$ 0,1,2,3,4,5,6,7 $$

Characteristics

1. It has base 8.
2. It uses eight unique digits.
3. Each position represents a power of 8.
4. One octal digit represents exactly 3 binary bits.
5. It provides a shorter representation of binary numbers.

Example

Consider:

$$ (157)_8 $$

Expand:

$$ =1\times8^2+5\times8^1+7\times8^0 $$
$$ =64+40+7 $$
$$ =(111)_{10} $$

6. Hexadecimal Number System (Base 16)

PPT - Hexadecimal Number System PowerPoint Presentation, free download - ID:4637686 slideserve.com

Definition:

The Hexadecimal Number System is a number system with base 16 that uses sixteen symbols, including digits 0–9 and letters A–F.

Digits Used

Symbol	Decimal Value
0–9	0–9
A	10
B	11
C	12
D	13
E	14
F	15

Characteristics

1. It has base 16.
2. It uses sixteen unique symbols.
3. It uses digits 0–9 and letters A–F.
4. Each position represents a power of 16.
5. One hexadecimal digit represents exactly 4 binary bits.

Example

Consider:

$$(2A)_{16}$$

Here, A = 10.

Expand:

$$=2\times16^1+10\times16^0$$
$$=32+10$$
$$=(42)_{10}$$

7. Comparison of All Four Number Systems

Important Theory Question

Feature	Decimal	Binary	Octal	Hexadecimal
Base	10	2	8	16
Digits	0–9	0–1	0–7	0–9, A–F
Number of symbols	10	2	8	16
Positional powers	10	2	8	16
Binary bits represented by one digit	Variable	1	3	4

Important conversion relationship:

$$1\ \text{Octal digit}=3\ \text{Binary bits}$$
$$1\ \text{Hexadecimal digit}=4\ \text{Binary bits}$$

PART 3: Positional Notation

8. What is Positional Notation?

Definition:

Positional notation is a number representation method in which the value of each digit depends on its position and the base of the number system.

Explanation

In a positional number system, the same digit can have different values depending on where it appears.

For example:

$$ 555 $$

The three 5s have different positional values:

- First 5 = 500
- Second 5 = 50
- Third 5 = 5

This happens because each position represents a different power of the base.

General Formula

For a number with base b:

$$ N=d_n b^n+d_{n-1}b^{n-1}+\cdots+d_1b^1+d_0b^0 $$

Where:

- N = Number
- b = Base
- d = Digit
- Position starts from 0 on the right.

Example 1: Decimal Positional Notation

Convert:

$$ (472)_{10} $$

Solution:

$$ =4\times10^2+7\times10^1+2\times10^0 $$
$$ =400+70+2 $$
$$ =472 $$

Example 2: Binary Positional Notation

Convert:

$$ (1101)_2 $$

Solution:

$$ =1\times2^3+1\times2^2+0\times2^1+1\times2^0 $$
$$ =8+4+0+1 $$
$$ =(13)_{10} $$

Example 3: Octal Positional Notation

Convert:

$$ (246)_8 $$

Solution:

$$ =2\times8^2+4\times8^1+6\times8^0 $$
$$ =128+32+6 $$
$$ =(166)_{10} $$

Example 4: Hexadecimal Positional Notation

Convert:

$$ (3B)_{16} $$

Remember:

$$ B=11 $$

Solution:

$$ =3\times16^1+11\times16^0 $$
$$ =48+11 $$
$$ =(59)_{10} $$

PART 4: Number Representation in Computers

9. How Are Numbers Represented in Computers?

Definition:

Computers represent numbers using binary digits (0 and 1), because their digital electronic circuits operate using two distinguishable states.

Explanation

Computers do not directly understand decimal digits such as 1, 2, 3, or 9.

Instead, decimal numbers are converted into binary form before processing.

For example:

\[(10)_{10}=(1010)_2 \]

The computer stores and processes the binary representation.

Why Is Binary Important?

1. Digital circuits work with two stable states.
2. Binary representation simplifies electronic circuit design.
3. It provides reliable data storage and processing.
4. It supports arithmetic and logical operations.
5. It is used to represent instructions, characters, and numerical values.

Important Concept

All data inside a digital computer is ultimately represented using binary patterns.

Examples:

- Numbers
- Text
- Images
- Audio
- Instructions

PART 5: Number-System Conversion Methods

This is the most important part of Week 3 for numerical questions.

There are two main approaches:

1. Positional Weight Method: Multiply each digit by its positional power and add the results.

2. Repeated Division Method: Divide the decimal number by the target base and read the remainders from bottom to top.

Let's learn each conversion step by step.

10. Decimal to Binary Conversion

Definition:

Decimal-to-binary conversion is the process of converting a base-10 number into its equivalent base-2 representation.

Method: Repeated Division by 2

Steps:

1. Divide the decimal number by 2.
2. Write down the remainder (0 or 1).
3. Divide the quotient by 2 again.
4. Repeat until the quotient becomes 0.
5. Read all remainders from bottom to top.

Example 1

Convert:

\[(25)_{10}=(?)_2 \]

Solution:

Repeated Division by 2

Division	Quotient	Remainder
25 ÷ 2	12	1
12 ÷ 2	6	0
6 ÷ 2	3	0
3 ÷ 2	1	1
1 ÷ 2	0	1

Read remainders from bottom to top:

11001

Final answer

$$\boxed{(25)_{10}=(11001)_2}$$

Example 2

Convert:

$$(45)_{10}=(?)_2$$

Division	Quotient	Remainder
45 ÷ 2	22	1
22 ÷ 2	11	0
11 ÷ 2	5	1
5 ÷ 2	2	1
2 ÷ 2	1	0
1 ÷ 2	0	1

Read from bottom to top:

$$\boxed{(45)_{10}=(101101)_2}$$

Example 3

Convert:

$$(100)_{10}=(?)_2$$

Division	Quotient	Remainder
100 ÷ 2	50	0
50 ÷ 2	25	0
25 ÷ 2	12	1
12 ÷ 2	6	0
6 ÷ 2	3	0
3 ÷ 2	1	1
1 ÷ 2	0	1

Reading bottom to top:

$$\boxed{(100)_{10}=(1100100)_2}$$

Remember: Always read the remainders from bottom to top, not from top to bottom.

11. Binary to Decimal Conversion

Definition:

Binary-to-decimal conversion is the process of converting a base-2 number into its equivalent base-10 representation.

Method: Positional Weight Method

Steps:

1. Write the binary number.
2. Assign powers of 2 from right to left, starting with 2^0.
3. Multiply each binary digit by its corresponding power.
4. Add all the results.

Example 1

Convert:

$$ (1011)_2=(?)_{10} $$

Solution:

Binary digit	**1**	**0**	**1**	**1**
Power	2^3	2^2	2^1	2^0
Value	8	0	2	1

Now add:

$$ 8+0+2+1=11 $$
$$ \boxed{(1011)_2=(11)_{10}} $$

Example 2

Convert:

$$ (110101)_2=(?)_{10} $$

Solution:

$$ =1\times2^5+1\times2^4+0\times2^3 $$
$$ +1\times2^2+0\times2^1+1\times2^0 $$
$$ =32+16+0+4+0+1 $$
$$ \boxed{(110101)_2=(53)_{10}} $$

Example 3

Convert:

$$ (1001101)_2=(?)_{10} $$

Solution:

$$ =1\times2^6+0\times2^5+0\times2^4 $$
$$ +1\times2^3+1\times2^2+0\times2^1+1\times2^0 $$
$$ =64+0+0+8+4+0+1 $$
$$ \boxed{(1001101)_2=(77)_{10}} $$

Important Powers of 2

Memorize this table:

Power	Value
2^0	1
2^1	2
2^2	4
2^3	8
2^4	16
2^5	32
2^6	64
2^7	128
2^8	256
2^9	512
2^{10}	1024

12. Binary to Hexadecimal Conversion

Definition:

Binary-to-hexadecimal conversion is the process of converting a binary number into its equivalent base-16 representation by grouping binary digits into sets of four.

Method: Grouping of Four Bits

Steps:

1. Start grouping binary digits from the right side.
2. Make groups of four bits.
3. Add leading zeros to the left if needed.
4. Convert each group into its hexadecimal equivalent.

5. Combine the hexadecimal digits.

Binary to Hexadecimal Table

Memorize this table

Binary	Hex	Binary	Hex
0000	0	1000	8
0001	1	1001	9
0010	2	1010	A
0011	3	1011	B
0100	4	1100	C
0101	5	1101	D
0110	6	1110	E
0111	7	1111	F

Example 1

Convert:

$$(10110110)_2=(?)_{16}$$

Step 1: Group into four bits.

$$1011\quad0110$$

Step 2: Convert each group.

$$1011=B$$
$$0110=6$$

Final answer:

$$\boxed{(10110110)_2=(B6)_{16}}$$

Example 2

Convert:

$$(110101)_2=(?)_{16}$$

Step 1: Group from the right.

$$11\quad0101$$

Add two zeros to the left:

$$ 0011\quad0101 $$

Step 2: Convert.

$$ 0011=3 $$
$$ 0101=5 $$
$$ \boxed{(110101)_2=(35)_{16}} $$

Example 3

Convert:

$$ (111100001010)_2=(?)_{16} $$

Groups:

$$ 1111\quad0000\quad1010 $$

Convert:

- 1111 = F
- 0000 = 0
- 1010 = A

$$ \boxed{(111100001010)_2=(F0A)_{16}} $$

13. Hexadecimal to Binary Conversion

Definition:

Hexadecimal-to-binary conversion is the process of converting each hexadecimal digit into its equivalent four-bit binary representation.

Method

1. Take each hexadecimal digit separately.
2. Find its binary equivalent using four bits.
3. Replace each hexadecimal digit with its binary equivalent.
4. Combine the binary groups.

Example 1

Convert:

$$ (A5)_{16}=(?)_2 $$

Step 1: Convert A.

$$ A=10=1010 $$

Step 2: Convert 5.

$$ 5=0101 $$

Step 3: Combine.

$$ \boxed{(A5)_{16}=(10100101)_2} $$

Example 2

Convert:

$$ (3F)_{16}=(?)_2 $$

- 3 = 0011
- F = 1111

Combine:

$$ \boxed{(3F)_{16}=(00111111)_2} $$

Example 3

Convert:

$$ (2B7)_{16}=(?)_2 $$

- 2 = 0010
- B = 1011
- 7 = 0111

Combine:

$$ \boxed{(2B7)_{16}=(001010110111)_2} $$

Important: Every hexadecimal digit must be represented using exactly four bits, including leading zeros.

PART 6: Octal Number-System Conversions

14. Binary to Octal Conversion

Definition:

Binary-to-octal conversion is the process of converting a binary number into its equivalent octal representation by grouping binary digits into sets of three.

Method

1. Start grouping from the right side.
2. Make groups of three bits.
3. Add leading zeros if necessary.
4. Convert each group into its octal equivalent.

Binary to Octal Table

Binary	Octal
000	0
001	1
010	2
011	3
100	4
101	5
110	6
111	7

Example 1

Convert:

$$ (110101)_2=(?)_8 $$

Group into three bits:

$$ 110\quad101 $$

Convert:

- 110 = 6
- 101 = 5

$$ \boxed{(110101)_2=(65)_8} $$

Example 2

Convert:

$$ (1011101)_2=(?)_8 $$

Group from right:

$$ 1\quad011\quad101 $$

Add leading zeros:

$$ 001\quad011\quad101 $$

Convert:

- 001 = 1
- 011 = 3
- 101 = 5

$$ \boxed{(1011101)_2=(135)_8} $$

15. Octal to Binary Conversion

Definition:

Octal-to-binary conversion is the process of converting each octal digit into its equivalent three-bit binary representation.

Example 1

Convert:

$$ (57)_8=(?)_2 $$

- 5 = 101
- 7 = 111

$$ \boxed{(57)_8=(101111)_2} $$

Example 2

Convert:

$$ (246)_8=(?)_2 $$

- 2 = 010
- 4 = 100
- 6 = 110

$$\boxed{(246)_8=(010100110)_2}$$

16. Decimal to Octal Conversion

Definition:

Decimal-to-octal conversion is the process of converting a base-10 number into its equivalent base-8 representation using repeated division by 8.

Example

Convert:

$$(125)_{10}=(?)_8$$

Divide by 8:

Division	Quotient	Remainder
125 ÷ 8	15	5
15 ÷ 8	1	7
1 ÷ 8	0	1

Read remainders from bottom to top:

$$\boxed{(125)_{10}=(175)_8}$$

17. Octal to Decimal Conversion

Definition:

Octal-to-decimal conversion is the process of converting a base-8 number into its equivalent base-10 representation using positional weights.

Example

Convert:

$$(157)_8=(?)_{10}$$

Solution:

$$ =1\times8^2+5\times8^1+7\times8^0 $$
$$ =64+40+7 $$
$$ \boxed{(157)_8=(111)_{10}} $$

PART 7: Additional Important Conversions

These conversions are useful because examiners can ask any two number systems even if the main syllabus emphasizes binary conversions.

18. Decimal to Hexadecimal Conversion

Definition:

Decimal-to-hexadecimal conversion converts a base-10 number into base 16 using repeated division by 16.

Example

Convert:

$$ (254)_{10}=(?)_{16} $$

Division	Quotient	Remainder
254 ÷ 16	15	14 (E)
15 ÷ 16	0	15 (F)

Read bottom to top:

$$ \boxed{(254)_{10}=(FE)_{16}} $$

19. Hexadecimal to Decimal Conversion

Definition:

Hexadecimal-to-decimal conversion converts a base-16 number into its equivalent base-10 value using positional weights.

Example

Convert:

$$ (2F)_{16}=(?)_{10} $$

Here:

$$ F=15 $$

Solution:

$$ =2\times16^1+15\times16^0 $$
$$ =32+15 $$
$$ \boxed{(2F)_{16}=(47)_{10}} $$

PART 8: Fractional Number Conversions

Your syllabus does not explicitly list fractional conversions, but understanding the basic method can help if the examiner gives a decimal point in a conversion question.

20. Decimal Fraction to Binary

For the fractional part, multiply repeatedly by 2.

- Take the integer part of each result.
- Continue multiplying the fractional part by 2.
- Read the integer parts from top to bottom.

Example

Convert:

$$ (0.625)_{10}=(?)_2 $$

Multiplication	Result	Integer bit
0.625 × 2	1.25	1
0.25 × 2	0.50	0
0.50 × 2	1.00	1

Read the integer bits from top to bottom:

$$ \boxed{(0.625)_{10}=(0.101)_2} $$

21. Binary Fraction to Decimal

Multiply each bit by its corresponding negative power of 2.

Example

Convert:

\[(0.101)_2=(?)_{10} \]

Solution:

\[=1\times2^{-1}+0\times2^{-2}+1\times2^{-3} \]
\[=0.5+0+0.125 \]
\[\boxed{(0.101)_2=(0.625)_{10}} \]

PART 9: Important Conceptual Questions

These questions are important for short questions, MCQs and conceptual explanations.

Q1. Why do computers use the binary number system?

Computers use the binary number system because their electronic circuits operate using two stable states, ON and OFF, represented by 1 and 0. Binary representation makes data processing and storage reliable and efficient.

Q2. Why is hexadecimal used in computer systems?

Hexadecimal provides a shorter and more readable representation of binary numbers. Each hexadecimal digit represents exactly four binary bits.

Q3. What is the difference between a bit and a byte?

A bit is the smallest unit of digital information and represents either 0 or 1. A byte consists of 8 bits.

Q4. What is radix?

Radix is another name for the base of a number system. It represents the total number of unique digits used in that system.

Q5. Why is the octal number system useful?

Octal provides a shorter representation of binary numbers because one octal digit represents three binary bits.

Q6. Can a binary number contain the digit 2?

No. Binary has base 2 and only uses digits 0 and 1.

Q7. Can an octal number contain the digit 8?

No. Octal has base 8 and uses digits from 0 to 7.

Q8. Can a hexadecimal number contain the letter G?

No. Hexadecimal uses digits 0–9 and letters A–F.

Q9. What is positional notation?

Positional notation is a number representation system in which the value of each digit depends on its position and the base of the number system.

Q10. What is the relationship between binary, octal and hexadecimal?

One octal digit represents three binary bits, while one hexadecimal digit represents four binary bits. Therefore, both systems provide compact representations of binary numbers.

PART 10: Mixed Conversion Practice

These examples help you understand how to move between different number systems.

Given Number	Conversion	Answer
\((15)_{10}\)	Decimal to Binary	\((1111)_2\)
\((20)_{10}\)	Decimal to Octal	\((24)_8\)
\((31)_{10}\)	Decimal to Hex	\((1F)_{16}\)
\((1111)_2\)	Binary to Decimal	\((15)_{10}\)
\((101010)_2\)	Binary to Decimal	\((42)_{10}\)
\((101010)_2\)	Binary to Octal	\((52)_8\)
\((11111111)_2\)	Binary to Hex	\((FF)_{16}\)
\((77)_8\)	Octal to Decimal	\((63)_{10}\)
\((3A)_{16}\)	Hex to Decimal	\((58)_{10}\)
\((64)_8\)	Octal to Binary	\((110100)_2\)

PART 11: Common Mistakes in Numericals

These mistakes can cause incorrect answers even if you understand the conversion method.

1. Decimal to Binary: Do not read remainders from top to bottom. Always read bottom to top.
2. Binary to Decimal: Start powers from \(2^0\) on the rightmost digit.
3. Binary to Hexadecimal: Group digits into four bits from the right.
4. Binary to Octal: Group digits into three bits from the right.

5. Hexadecimal: Remember A = 10, B = 11, C = 12, D = 13, E = 14 and F = 15.
6. Octal: Never use digits 8 or 9.
7. Binary: Never use digits other than 0 and 1.
8. Leading Zeros: Add zeros only on the left when completing binary groups. They do not change the number's value.

PART 12: Important Exam Questions

Short Questions

1. Define a number system.
2. What is radix?
3. Define binary number system.
4. What is the base of decimal, binary, octal and hexadecimal?
5. Why do computers use binary?
6. Define positional notation.
7. What is a bit, nibble and byte?
8. How many binary bits are represented by one octal digit?
9. How many binary bits are represented by one hexadecimal digit?
10. What are the hexadecimal symbols?
11. Why is hexadecimal useful in computer systems?
12. Differentiate between binary and decimal number systems.

Long Questions

1. Explain the four major number systems with examples.
2. Explain positional notation with suitable examples.
3. Explain how numbers are represented inside computers.
4. Explain decimal-to-binary and binary-to-decimal conversions with examples.
5. Explain binary-to-hexadecimal and hexadecimal-to-binary conversions.
6. Explain octal-to-binary and binary-to-octal conversions.
7. Explain different number-system conversion techniques with solved examples.

PART 13: Numerical Practice Test

Try these questions yourself. These cover the main conversion patterns from your syllabus.

Week 3 — MCQ Test

0/10 attempted

Q1. Convert $(35)_{10}$ into binary.

100011
100101
101011
110001

Q2. Convert $(1101)_2$ into decimal.

11
12
13
14

Q3. Convert $(10101100)_2$ into hexadecimal.

AC
BC
AB
CA

Q4. Convert $(3F)_{16}$ into binary.

00111111
00110111
11110011
11001111

Q5. Convert $(110101)_2$ into octal.

63
65
53
55

Q6. Convert $(157)_8$ into decimal.

109
110
111
112

Q7. Convert $(254)_{10}$ into hexadecimal.

EF
FE
FD
FF

Q8. What is the base of hexadecimal?

2
8
10
16

Q9. How many binary bits are represented by one octal digit?

2
3
4
8

Q10. Which digits are allowed in the binary system?

0–7
0–9
0 and 1
0–F
Submit Answers

WEEK 03 — Final Revision Sheet

1. Number System Bases

System	Base
Binary	2
Octal	8
Decimal	10
Hexadecimal	16

2. Important Conversion Rules

- Decimal → Binary: Divide by 2.
- Binary → Decimal: Multiply by powers of 2.
- Binary → Hexadecimal: Group 4 bits.
- Hexadecimal → Binary: Replace each digit with 4 bits.
- Binary → Octal: Group 3 bits.
- Octal → Binary: Replace each digit with 3 bits.
- Decimal → Octal: Divide by 8.
- Decimal → Hexadecimal: Divide by 16.

3. Important Powers

Power	Value
2^0	1
2^1	2
2^2	4
2^3	8
2^4	16
2^5	32
8^0	1
8^1	8
8^2	64
16^0	1
16^1	16

My recommendation for your midterm

Give maximum practice to these six numerical patterns:

1. Decimal to Binary using repeated division.
2. Binary to Decimal using positional weights.
3. Binary to Hexadecimal using four-bit grouping.
4. Hexadecimal to Binary using the conversion table.
5. Binary to Octal using three-bit grouping.
6. Octal to Decimal using positional weights.

One important exam tip: Do not just memorize the conversion tables. Practice at least two examples of every conversion method. If the examiner changes the numbers, you should still be able to solve them.

Next is Week 04 — Data Representation, Signed and Unsigned Integers, 1's Complement, 2's Complement, Overflow, Fixed-Point and Floating-Point Representation, IEEE 754, ASCII and Unicode. This week is especially important for numerical questions involving signed binary numbers and floating-point representation.